Higher National Diploma in Engineering (Civil)

Second Year, Second Semester Examination - 2018

CE 2212-Structural Design

Q1.

A 300mm square column carries a dead load of 1200kN and imposed load of 360kN. The safe bearing capacity of the soil is 190 kN/m^2, Design a reinforced concrete square pad footing to resist the above loads. The characteristic strength of concrete and yield strength of steel are 35 N/mm^2 and 460 N/mm^2 respectively. Carry out the relevant checks also.

(Total 20 Marks)

Q2

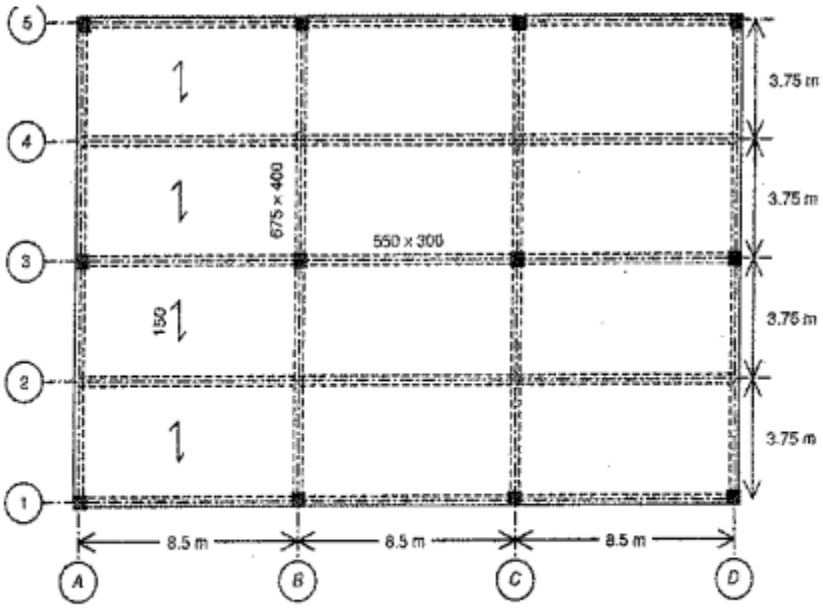

Fig Q2

The above shown fig Q2 shows the arrangements of slab, columns and beams. Design the **main reinforcement to slab BC23** in this arrangement by using BS 8110. Perform the **crack control and deflection checks** assume the values for Finishes as 1.5kN/m^2 and imposed load of 4kN/m^2. Consider that f_{cu} = 35 N/mm^2, exposure condition mild and fire resistance of 1 hour.

(Total 20 Marks)

Q3:

An internal column in a braced two storey building, supporting an approximately symmetrical arrangement of beams which are 350mm wide X 600mm deep, results in characteristic dead load of 1200kN and imposed load of 1000kN being applied to the column. The column is 350mm square and has a clear height of 4.5m. Assume that the bottom side of this column is resisting on a pad foundation which is not designed to resist a moment. Design the longitudinal reinforcement and links. F_{cu} = 40N/mm^2 and f_y and f_{yv} are 500N/mm^2.

(Total 20 Marks)

Q4:

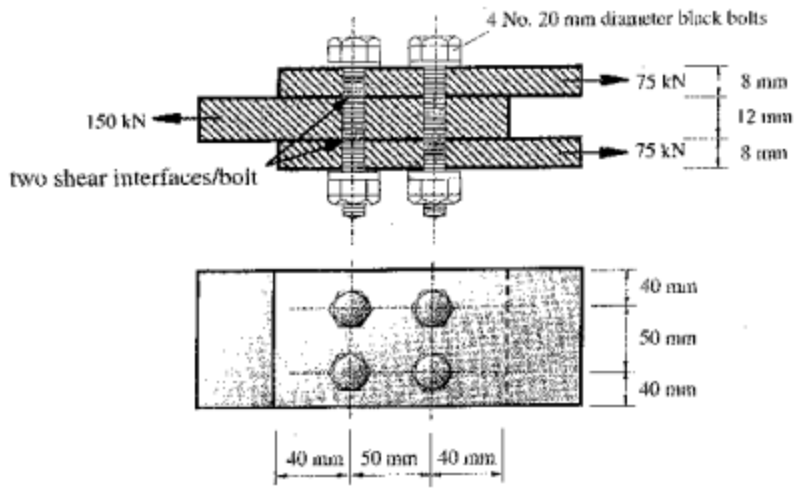

Figure Q4

A double lap joint is shown in the figure Q4. There are two shear interfaces and four 20mm diameter grade 8.8 black bolts. The outer plates are 8mm thick and the inner plate is 12mm thick. All are S275 steel. Check the spacing, minimum and maximum end and edge distances and check the bolt shear, bolt bearing, plate bearing, tension strength of bolts and plates.

(Total 20 Marks)

Q5.

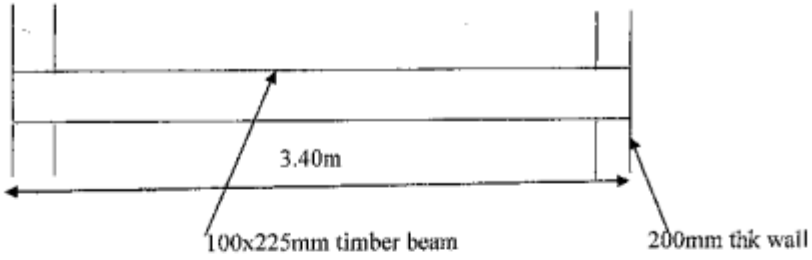

Fig Q5

A timber beam, total length of 3.40m spans over an opening and supports on 200mm thick brick wall at the ends (as shown in fig Q5). The beam is subjected to long duration loading of 3kN/m uniformly distributed design load. Check whether 100x225mm teak (D70) timber section is satisfied at wet exposure condition. *(Total 20 Marks)*

Q6.

i. Mention two factors affected for the characteristic compressive strength of brick work(f_k) (02 marks)

ii. A two story building has a roof slab of 100mm thick cast in situ one way spanning (4m span). Floor to floor has clear height is 2.85m of each stories. The dead load of finishes and services etc of the roof slab is 0.75kN/m^2 and imposed load on roof is 0.25kN/m^2. Calculate the thickness of the typical internal brick wall in the first floor level assuming that,

Effective height of the wall = 0.75 X clear height of the wall
The partial factor of safety for normal manufacturing control and normal construction control, $\gamma m = 3.5$
Characteristic compressive strength of masonry, f_k= 2.5N/mm2
Neglect lateral load effects
Design strength = $\beta b t f_k/\gamma m$ (18 marks)

(Total 20 Marks)

Bolt Grade	Shear strength , p_s (N/mm^2)
4.6	160
8.8	375
10.9	400

Bearing Strength of Bolts

Bolt Grade	Bearing Strength
4.6	460
8.8	1000
10.9	1300

Bearing Strength of Connected plates

Steel Grade	S275	S355	S460
Bearing Strength(N/mm^2)	460	550	670

Tensile stress area

Nominal size and tread diameter (mm)	Tensile stress area, A_s (mm^2)
15	84.5
16	157
20	245
22	303
24	353

Tensile strength of bolts

Bolt grade	Tensile strength p_t (N/mm^2)
4.6	240
8.8	560
10.9	700

Design strength for steel grade S275

Thickness (mm) less than or equal to	16	40	63	80
Design strength, p_y (N/mm^2)	275	265	255	245

Bearing Capacity of bolt, $P_{bb} = dt_b p_{bb}$
Bearing capacity of connected plates, $P_{bs} = k_{bs} d t_p p_{bs}$ but $P_{bs} \leq 0.5 k_{bs} e t_p p_{bs}$
$k_{bs} = 1$ for Standard clearance holes
e = End distance
d = Nominal diameter of bolt
t_p = Thickness of the connected plates
Tension capacity of bolt , $P_{tem} = 0.8 p_t A_t$
Tension capacity of plate, $P_t = p_y A_e$
A_e- Effective Area, $A_e = K_e A_n$
A_n- Net area
K_e – 1.2 for S275

Cross-sectional areas of groups of bars (mm^2)

Bar Size	Number of Bars * Denotes non-preferred sizes									
	1	2	3	4	5	6	7	8	9	10
6mm	28.3	56.5	84.8	113	141	170	198	226	255	283
8mm	50.3	100	151	201	251	302	352	402	452	503
10mm	78.5	157	236	314	339	471	550	628	707	785
12mm	113.1	226	339	452	566	679	792	905	1718	1131
16mm	201.1	402	603	804	1005	1206	1407	1608	1810	2011
20mm	314.2	628	942	1257	1571	1885	2199	2513	2827	3142
25mm	490.9	982	1473	1963	2454	2945	3436	3927	4418	4909
32mm	804.2	1608	2413	3217	4021	4825	5630	6434	7238	8042
40mm	1257.0	2513	3770	5026	6283	7540	8790	10053	11310	12566
50mm*	1963.5	3927	5890	7854	9812	11781	13744	15708	17671	19635

Cross-sectional areas for Specific Spacings in mm² / Metre Width

Bar Size	Spacing of Bars (Millimetres) * Denotes non-preferred sizes									
	75	100	125	150	175	200	225	250	275	300
6mm *	377	283	226	188	162	141	125	113	103	94
8mm	670	503	402	335	287	251	223	201	183	168
10mm	1047	786	628	524	449	393	348	314	285	262
12mm	1508	1131	905	754	646	565	503	452	411	377
16mm	2681	2011	1608	1340	1149	1005	894	801	731	670
20mm	4189	3142	2513	2094	1795	1571	1396	1257	1143	1047
25mm	6545	4909	3927	3272	2805	2454	2182	1963	1785	1636
32mm	-	8042	6434	5362	4596	4021	3574	3217	2924	2681
40mm	-	-	10053	8378	7181	6283	5587	5027	4571	4189
50mm*	-	-	-	13090	11220	9617	8727	7854	7140	6545

Capacity reduction factor, β

Slenderness ratio	Eccentricity at top of the wall , e			
	Upto 0.05t	0.1t	0.2t	0.3t
0	1.00	0.88	0.66	0.44
6	1.00	0.88	0.66	0.44
8	1.00	0.88	0.66	0.44
10	0.97	0.88	0.66	0.44
12	0.93	0.87	0.66	0,44